ESTAS-AQUABC Tutorial: Getting Started

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Contents

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A step-by-step practical guide to running the ESTAS-AQUABC ecological model using the Shiny web interface

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Table of Contents

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1. Overview

ESTAS-AQUABC is a coupled hydrodynamic–biogeochemical model for simulating water quality in aquatic systems. The model couples:

- **ESTAS** — a finite-element hydrodynamic transport solver (25-box Curonian Lagoon domain, advective + dispersive transport)
- **AQUABC** — an ecological/biogeochemical module with 36 state variables and 318+ calibratable parameters covering phytoplankton, zooplankton, nutrients, dissolved oxygen, carbon cycling, metals, and sediment fluxes

This tutorial uses the **Python Shiny web interface** to walk you through a complete model run using the pre-configured data files shipped with the repository. By the end you will have:

- Built the Fortran executable from source
- Configured and executed a 30-day simulation

- Inspected model parameters and initial conditions
- Plotted output variables and input forcing timeseries
- Evaluated mass balance results
- Compared model output against observations

What data files are used?

All input data is already present in the `INPUTS/` directory (89 files). The tutorial uses `INPUT_30day.txt` — a lightweight 30-day configuration (days 6209–6239, year 1998) that completes in a few minutes on a modern machine. The full 365-day configuration (`INPUT.txt`) is also available for later exploration.

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2. Prerequisites

Requirement	Details
Operating System	Linux (tested on Ubuntu 22.04+)
Fortran Compiler	`gfortran` (GCC 9+) or Intel `ifort/ifx`
Python	3.10 or newer
Python packages	`shiny`, `plotly`, `pandas`, `numpy`, `openpyxl`, `ipywidgets`, `shinywidgets`, `fpdf2` (see `shiny_app/requirements.txt`)
Make	GNU Make
Browser	Any modern browser (Chrome, Firefox, Edge)

Quick environment setup

```
# Navigate to the project root
cd /path/to/AQUABCv0.2

# Create a virtual environment (one-time)
python -m venv .venv
source .venv/bin/activate

# Install Python dependencies
pip install -r shiny_app/requirements.txt

# Verify gfortran is available
gfortran --version
```

If you are using the pre-configured server deployment, the environment is already set up at `/opt/micromamba/envs/shiny/` and the app runs as a systemd service.

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3. Starting the Application

Option A: Development mode (recommended for this tutorial)

```
cd /path/to/AQUABCv0.2
source .venv/bin/activate
shiny run --reload shiny_app:app
```

Open your browser to **http://127.0.0.1:8000**.

Option B: Production deployment

If deployed via `shiny_app/deploy.sh`, the app is accessible on the machine's HTTP address (port 80 via nginx reverse proxy). Check status:

```
sudo systemctl status shiny_aquabc
```

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4. Application Layout

The application has a **sidebar navigation** on the left and a **main content area** on the right. The header bar at the top provides access to the **Changelog** (journal icon), **State Variables Reference** (question mark icon), and **Settings** (gear icon).

Navigation Pages

#	Page	Purpose
1	**Dashboard**	Quick Run button, system status, simulation config summary, run log
2	**Model Structure**	Interactive network diagram of all state variables and interactions
3	**Model Build**	Compile the Fortran source code (gfortran/ifort, release/debug/fast)
4	**Model Config**	Configure simulation parameters, run the model, set output options
5	**Input Files**	Browse, preview, and analyse all files in `INPUTS/` with map display
6	**Parameters**	View and edit the 318 model constants in `WCONST_04.txt` by category
7	**Initial Cond.**	View and edit initial concentrations for all 36 state variables
8	**Model Options**	Toggle model switches (zooplankton, redox, light, cyanobacteria, etc.)
9	**Scenarios**	Save/load named scenario presets (parameters + ICs + options)
10	**Plots**	Visualise model output, input timeseries, and data previews
11	**Mass Balance**	Calculate and display element mass balance (N, C, P, Si)

#	Page	Purpose
12	**Observations**	Load observation data and compare against model output
13	**Map**	Geographic 3D map visualization of the model domain

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5. Tutorial Exercise: 30-Day Simulation

This exercise walks through every major page of the application using the bundled Curonian Lagoon dataset.

Step 1: Dashboard Overview

1. **Click "Dashboard"** in the sidebar (it is selected by default on startup).

2. You will see three main sections:

   - **Quick Run / Stop buttons** and a run timer at the top
   - **System Status** (left) — shows OS, Python version, available executables, current model configuration, and the command that will be executed
   - **Simulation Config** (centre) — displays key parameters loaded from the selected `INPUT*.txt` file (base year, time period, steps/day, folders)
   - **Run Log** (right) — shows real-time output when a simulation is running

3. Review the **System Status** panel. It shows:

   - The selected executable (default: `ESTAS_II`)
   - The selected input file (default: `INPUT.txt`)
   - The number of input files found in `INPUTS/`

4. **Do not click Quick Run yet** — we will first build the executable and configure the simulation.

   Tip: The "Model Config" button below the System Status panel is a shortcut to the Model Config page.

Step 2: Building the Model

Before running a simulation, you need a compiled executable.

1. **Click "Model Build"** in the sidebar.

2. The page has three columns:

   - **Build Configuration** (left) — compiler, build type, and options
   - **Available Executables** (centre) — list of compiled executables
   - **Build Log** (right) — compilation output

3. **Configure the build:**

- **Compiler:** Select `gfortran (GNU)` (default, recommended)
- **Build Type:** Select `Release` (optimised, good for production runs)
- Leave **"Clean before build"** unchecked for a normal build
- The **Target Executable** will show `ESTAS_gf_release`

4. **Click the "Build" button.**
5. Watch the **Build Log** panel on the right. The compilation takes approximately 30–60 seconds depending on your machine. You will see:
   - Individual Fortran source files being compiled
   - Library archive creation
   - Final linking step
   - A success or failure message
6. Once the build completes successfully:
   - The new executable appears in the **Available Executables** list
   - Use the **"Select for Run"** dropdown to choose the newly built executable
   - If `ESTAS_II` already exists (pre-compiled), you may use that instead

   Note: If you already have a compiled `ESTAS_II` executable (shown in the Available Executables list), you can skip building and proceed directly to **Step 3**. The repository includes a pre-compiled `ESTAS_II` (gfortran, 64-bit Linux).

Step 3: Exploring Input Files

1. **Click "Input Files"** in the sidebar.
2. The page has two main sections:
   - **File Browser** (left) — file list with category filter and file information
   - **File Contents / Map Display** (right) — preview panel with two tabs
3. **Use the category filter** dropdown to explore files by type:
   - *Forcing Data* — meteorological and hydrological timeseries
   - *Geometry* — box connectivity and bathymetry
   - *Initial Conditions* — starting concentrations
   - *Constants* — model parameter files
   - *Sediment* — prescribed sediment flux files
4. **Select and preview key files:**

File	Description	What to look for
`PELAGIC_INPUTS.txt`	Master pelagic config	Box count, layer structure, file references
`ADVECTIVE_LINKS.txt`	Box connectivity	Transport link definitions (from → to)
`TEMP_TS.txt`	Water temperature	Time-varying forcing for all 25 boxes
`FLOW_TS.txt`	Water flow rates	Hydrodynamic forcing timeseries
`WCONST_04.txt`	Model constants	All 323 calibrated parameter values
`INIT_CONC_1.txt`	Initial conditions	Starting concentrations for 36 variables

5. **Check the File Information** panel below the file list — it shows file size, line count, detected format, and data range for timeseries files.

6. **Switch to the "Map Display" tab** to see:

   - **Box Network** — connectivity diagram of the 25 model boxes
   - **Bathymetry Profile** — depth profile for a selected box
   - **Box Depths Overview** — comparative depth view across all boxes

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Step 4: Reviewing Parameters

1. **Click "Parameters"** in the sidebar.

2. The page shows parameter values from the constants file (`WCONST_04.txt`), organised into 14 categories:

Category	Parameters	Description
General	1–17	Light, temperature functions, global switches
Diatoms	18–54	Diatom growth, respiration, mortality
Non-fixing Cyanobacteria	55–89	Non-N2-fixing cyano processes
Fixing Cyanobacteria	90–124	N2-fixing cyanobacteria processes
Other Phytoplankton	125–159	Other algae (OPA) kinetics
Zooplankton	160–192	Grazing, respiration, mortality
Detritus	193–218	Particulate organic matter
Dissolved Organics	219–234	DOM cycling
Nitrification	235–250	NH4 → NO3 transformation
Redox Chemistry	251–272	Fe, Mn, S redox
Methane	273–283	CH4 production and oxidation
Settling	284–297	Particle settling velocities
pH Effects	298–305	Carbonate system, alkalinity
Nostocales	306–323	Nostocales-specific parameters

3. **Select a category** (e.g., "Diatoms") and click **"Load"**.

4. The **Parameters** table shows:

   - Parameter index number
   - Current value
   - An editable input field for each parameter

5. **For this tutorial, do not modify any parameters.** Simply review the values to familiarise yourself with the model calibration. The supplied `WCONST_04.txt` contains the latest calibrated parameter set.

   Tip: If you do modify parameters, click **"Save All Changes"** to write the updated values back to the file. A `.bak` backup is created automatically before any save.

Step 5: Setting Initial Conditions

1. **Click "Initial Cond."** in the sidebar.
2. Select the IC file: `INIT_CONC_1.txt` (default).
3. Choose a category to view. State variables are grouped into 11 categories:

Category	Variables
Nutrients	NH4-N, NO3-N, PO4-P, Dissolved Si
Dissolved Gases	Dissolved Oxygen
Phytoplankton	DIA_C, CYN_C, OPA_C, FIX_CYN_C, NOST_VEG_HET_C, AKI_C
Zooplankton	ZOO_C, ZOO_N, ZOO_P
Particulate Organics	DET_PART_ORG_C, DET_PART_ORG_N, DET_PART_ORG_P, PART_Si
Dissolved Organics	DISS_ORG_C, DISS_ORG_N, DISS_ORG_P
Carbonate System	INORG_C, TOT_ALK
Metals	FE_II, FE_III, MN_II, MN_IV, CA, MG
Sulphur	S_PLUS_6, S_MINUS_2
Allelopathy	SEC_METAB_DIA, SEC_METAB_NOFIX_CYN, SEC_METAB_FIX_CYN, SEC_METAB_NOST
Other	CH4_C

4. Click **"Load"** to display the current initial values.
5. Review the concentrations. The values represent starting conditions at simulation time 6209 (approximately 1 January of the simulation year).
6. **For this tutorial, keep the default values and proceed.**

Step 6: Configuring Model Options

1. **Click "Model Options"** in the sidebar.
2. Select a category from the dropdown. Available categories:

Category	What it controls
Zooplankton	Feeding preference, switching function selection
Redox Chemistry	Fe/Mn/S cycling switches
Light	Light limitation formulation
Cyanobacteria	N2 fixation switches, CTMI temperature model
Allelopathy	Secondary metabolite interactions
Organic Matter	DOM/POM processing options

3. Click **"Load Options"** to display the current settings.

4. The page shows two panels:
   - **Model Switches** — on/off toggles for various model features
   - **Extra Constants** — additional constants from `EXTRA_WCONST.txt`
5. **For this tutorial, keep the default options and proceed.**

Step 7: Configuring the Simulation

1. **Click "Model Config"** in the sidebar.
2. The page has three tabs: **Simulation Config**, **Run Model**, and **Output Config**.

Tab 1: Simulation Config

3. Click **"Load Configuration"** to read the current `INPUT.txt` settings.
4. Configure the 30-day tutorial run:

Setting	Value	Notes
Base Year	`1998`	Reference year for forcing data
Start Date	Corresponds to day 6209	~1 Jan of the simulation year
End Date	Corresponds to day 6239	~31 Jan (30-day run)
Time Step	`6 minutes` (240 steps/day)	Recommended default
Output Frequency	`Hourly` (print interval = 10)	10 × 6 min = 1 hour
Enable Sediment Model	`Off`	Faster run without sediment diagenesis
Resuspension Option	`0 (Disabled)`	Simplest configuration for tutorial

   Alternatively — and **recommended for this tutorial** — you can simply select `INPUT_30day.txt` as the input file in the Run Model tab (next step), which already has these settings pre-configured.

5. If you modified settings, click **"Save Configuration"** to write them to the input file.

Tab 2: Run Model

6. Switch to the **"Run Model"** tab.
7. Configure the run parameters:
   - **Executable:** Select the executable you built in Step 2 (e.g., `ESTAS_gf_release`) or use the default `ESTAS_II`
   - **Input Configuration File:** Select `INPUT_30day.txt` from the dropdown — this is the pre-configured 30-day tutorial run
   - **Pelagic Constants File:** Select `WCONST_04.txt` (recommended) or leave as "(not used - use defaults)" to use built-in constants
   - **Enable Binary Output:** Leave `Off` for this tutorial

8. Check the **Command Preview** box at the bottom of the left panel. It should show something like:

   ```
   ./ESTAS_II INPUT_30day.txt WCONST_04.txt
   ```

   If using a custom-named executable:

   ```
   ./ESTAS_gf_release INPUT_30day.txt WCONST_04.txt
   ```

Tab 3: Output Config

9. Switch to the **"Output Config"** tab (optional but informative).
10. Review the **Output Boxes** selection. By default, boxes 5, 6, 8, 9, 14, 17, and 25 are selected. These are the boxes for which output files (`PELAGIC_BOX_*.out`) will be written.
11. Review the **Output Directory**. For `INPUT_30day.txt`, output goes to `OUTPUTS_30day/`.
12. The **Output Types** section lets you select:
    - **State Variables** — concentration timeseries (always recommended)
    - **Process Rates** — biogeochemical flux timeseries
    - **Mass Balance** — conservation check timeseries

Step 8: Running the Simulation

You can run the simulation from either the **Dashboard** or the **Model Config → Run Model** tab.

Method A: Quick Run from Dashboard

1. Navigate to the **Dashboard**.
2. Verify the System Status panel shows the correct executable and input file.
3. Click the green **"Quick Run"** button.

Method B: Run from Model Config

1. Navigate to **Model Config → Run Model** tab.
2. Verify the command preview is correct.
3. Click the green **"Run Model"** button.

Monitoring the Run

4. The **Run Log** panel (right side of Dashboard, or right side of Run Model tab) shows real-time output:

   ```
   Starting quick run...
   Validating input files...
   OK Input files validated
   OK Constants file validated: WCONST_04.txt (323 constants)

   Command: ./ESTAS_II INPUT_30day.txt WCONST_04.txt
   ------------------------------------
   ```

```
Starting model execution...
```

5. The model will output progress as it processes each simulation day. A 30-day run typically completes in **2–5 minutes** depending on hardware.

6. On the Dashboard, a **timer** display shows elapsed time.

7. When the run finishes, the log shows:

   ```
   OK Model completed successfully
   ```

8. To **stop a running simulation**, click the red **"Stop"** button.

   What if the run fails? Check the Run Log for error messages. Common issues: - "Executable not found" → Go to Model Build and compile first - "Missing required file" → Check that `INPUTS/` directory has all files - "Constants file validation failed" → Select `WCONST_04.txt` which has all 323 required constants

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Step 9: Viewing Results

1. **Click "Plots"** in the sidebar.

2. The Plots page has four tabs:

   - **Output Directory** — select which output folder to analyse
   - **Model Output** — plot state variable timeseries
   - **Input Timeseries** — visualise forcing data
   - **Data Preview** — tabular view of raw data

Selecting the Output Directory

3. In the **Output Directory** tab:
   - Select `OUTPUTS_30day` from the dropdown (or click "Refresh Directories" if it's not listed)
   - Click **"Analyze Directory"** to see a summary of available output files

Plotting Model Output

4. Switch to the **Model Output** tab.

5. Configure the plot:

   - **File format:** Select `Text (.out)`
   - **Output file:** Select a box file, e.g., `PELAGIC_BOX_00005.out` (Box 5 — central lagoon area)

6. Select variables to plot:

   - **Left axis:** Choose one or more variables, e.g.:
     - `DISS_OXYGEN` — dissolved oxygen
     - `NH4_N` — ammonium nitrogen
   - **Right axis:** Optionally add a variable on a different scale, e.g.:
     - `DIA_C` — diatom carbon biomass

7. Set plot options:

- **Apply rolling mean:** Enable and set window size (e.g., 5) to smooth noisy output
- **Log scale:** Enable for variables spanning several orders of magnitude

8. Click **"Refresh Plot"** or the plot updates automatically.

9. The interactive Plotly chart allows:

   - **Hover** over data points to see exact values and time
   - **Zoom** by clicking and dragging to select a region
   - **Pan** by holding shift and dragging
   - **Reset zoom** by double-clicking
   - **Download** the plot as PNG using the camera icon in the toolbar

Suggested Variables to Explore

Variable	Unit	What it shows
`NH4_N`	mg N/L	Ammonium — key nutrient
`NO3_N`	mg N/L	Nitrate — nutrient, nitrification product
`PO4_P`	mg P/L	Phosphate — limiting nutrient
`DISS_OXYGEN`	mg O2/L	Dissolved oxygen — ecosystem health indicator
`DIA_C`	mg C/L	Diatom biomass
`CYN_C`	mg C/L	Non-fixing cyanobacteria biomass
`FIX_CYN_C`	mg C/L	Nitrogen-fixing cyanobacteria biomass
`ZOO_C`	mg C/L	Zooplankton biomass
`INORG_C`	mg C/L	Dissolved inorganic carbon
`TOT_ALK`	meq/L	Total alkalinity

Plotting Input Timeseries

10. Switch to the **Input Timeseries** tab.

11. Select a forcing file:

    - `Temperature` — water temperature (°C)
    - `Salinity` — water salinity (PSU)
    - `Flow` — water flow rates (m^3/s)
    - `Solar Radiation` — incoming solar radiation (W/m^2)
    - `Wind Speed` — wind speed (m/s)
    - `Air Temperature` — air temperature (°C)
    - `Shear Stress` — bottom shear stress (Pa)

12. Select boxes to display (e.g., "5", "14", "25" to compare lake regions).

13. Click **"Plot Timeseries"** to generate the chart.

Data Preview

14. The **Data Preview** tab shows raw tabular data from the currently loaded output file. Use this to inspect exact values.

Step 10: Analysing Mass Balance

1. **Click "Mass Balance"** in the sidebar.
2. Click the **"Calculate Mass Balance"** button.
3. The page shows:
    - **Summary table** — mass balance for Nitrogen, Carbon, Phosphorus, and Silicon showing initial mass, final mass, and relative error
    - **Element Details** — select an element (e.g., Nitrogen) to see a breakdown of all contributing state variables
    - **Time Series** — temporal evolution of total element mass over the simulation period
4. A well-calibrated run should show mass balance relative errors below 1%. Larger errors may indicate numerical instability or configuration issues.

Step 11: Comparing with Observations

1. **Click "Observations"** in the sidebar.
2. The page provides two ways to load observation data:

    Method A: From OBSERVATIONS directory

    - Click **"Scan OBSERVATIONS Directory"** to find available files
    - The bundled files include:
        - `Water_column_chemistry_2015.xlsx` — field measurements
        - `2015ST14.xlsx` — station 14 observation data
        - `PelagicProcesses.xlsx` — measured pelagic process rates
        - `.dates` files — date-indexed observation records
    - Select a file and click **"Load Selected File"**

    Method B: Upload your own

    - Use the file upload widget to load CSV or Excel files

    Method C: Generate synthetic data

    - Click **"Generate Sample Data"** to create synthetic observations for testing the comparison workflow
3. After loading observations:
    - **File Preview** shows the data structure and available columns
    - **Comparison Summary** shows statistical metrics (RMSE, R^2, bias) for matched variables
    - **Variable Details** — select a variable for detailed comparison metrics
    - **Scatter Plot Info** — model vs. observation scatter analysis

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6. Working with Scenarios

The **Scenarios** page lets you save and restore complete model configurations.

Saving a Scenario

1. **Click “Scenarios”** in the sidebar.
2. In the **“Save Current Configuration”** panel:
   - Enter a **name** (e.g., “Tutorial_baseline”)
   - Enter a **description** (e.g., “Default 30-day run with WCONST_04”)
   - Select what to include:
     ☒ Parameters (WCONST_04.txt)
     ☒ Initial Conditions (select INIT_CONC_1.txt)
     ☒ Model Options & Constants
   - Click **“Save as New Scenario”**

Loading a Scenario

3. Use the **“Select Scenario”** dropdown to choose a previously saved configuration.
4. Click **“Load”** to replace the current configuration with the saved one.
5. A scenario can be **deleted** using the “Delete” button if no longer needed.

   Why use scenarios? Scenarios are particularly useful for: - Saving a calibrated baseline before making changes - Comparing different parameter sets - Quickly switching between configurations for sensitivity analysis

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7. Geographic Visualization

1. **Click “Map”** in the sidebar.
2. The map shows the Curonian Lagoon domain centred at approximately 55.32°N, 21.10°E.
3. **Map controls:**
   - **Map Style:** Choose from OpenStreetMap, Light/Dark (Carto), Satellite (Esri), or Topographic
   - **Centre Latitude/Longitude:** Adjust map centre coordinates
   - **Zoom Level:** 1 (world) to 18 (street level); default 10
   - **Pitch (3D tilt):** Angle for 3D perspective view (default 45°)
   - **Point Radius:** Size of data points on the map
   - **Elevation Scale:** Vertical exaggeration for 3D visualisation
4. The map also displays sample data points corresponding to model box locations.

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8. Tips and Troubleshooting

Performance

- The **30-day simulation** (`INPUT_30day.txt`) is recommended for learning and testing. It completes in 2–5 minutes.

- The **365-day simulation** (`INPUT.txt`) takes 30–60 minutes and produces ~8,700 output lines per box.
- Use **Release** build type for production runs. Debug builds are 5–10× slower but provide better error diagnostics.

Common Issues

Problem	Solution
"Executable not found"	Go to Model Build and compile, or verify ESTAS_II exists
"Missing required file"	Check `INPUTS/` directory has PELAGIC_INPUTS.txt, PELAGIC_MODEL_OPTIONS.txt, TEMP_TS.txt, FLOW_TS.txt, ADVECTIVE_LINKS.txt, INIT_CONC_1.txt
"Constants file validation failed"	Use `WCONST_04.txt` which has all 323 required constants
Run produces NaN or crashes	Try Debug build type to get detailed error output; check initial conditions for zero or negative values
"Intel runtime libraries not found"	Only relevant for ifort-compiled executables; use gfortran instead
Plot shows no data	Verify the output directory matches the INPUT file configuration; click "Refresh Files"
App won't start	Check Python environment: `pip install -r shiny_app/requirements.txt`

Navigating Efficiently

- Use the **Dashboard** for a quick overview and one-click model run
- The **Model Structure** diagram provides a visual reference of all biogeochemical interactions — useful when interpreting output variables
- The **State Variables Reference** (question mark icon in header) provides a quick-reference table of all 36 state variables
- The **Settings** panel (gear icon) allows theme customisation

File Safety

- All parameter and IC file edits create automatic `.bak` backups
- The app writes directly to `INPUTS/` files — use Scenarios to save checkpoints before major changes
- Output files are overwritten on each new run into the configured directory

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9. Data Files Reference

Input Configuration Files (`INPUT*.txt`)

These files sit in the project root and control the overall simulation.

File	Duration	Resuspension	Sediment Model	Output Folder	Notes
`INPUT_30day.txt`	30 days	Disabled (0)	Off	`OUTPUTS_30day/`	**Tutorial — recommended**
`INPUT.txt`	365 days	Semi-prescribed (2)	Off	`OUTPUTS/`	Full annual simulation
`INPUT_gf_release.txt`	365 days	—	—	—	Release build config
`INPUT_gf_debug.txt`	365 days	—	—	—	Debug build config
`INPUT_gf_fast.txt`	365 days	—	—	—	Fast build config
`INPUT_debug_run.txt`	—	—	—	`OUTPUTS_debug_run/`	Debugging configuration
`INPUT_sediment_test.txt`	—	—	On	—	Sediment model testing

Key Input Data Files (`INPUTS/`)

File(s)	Category	Description
`PELAGIC_INPUTS.txt`	Core	Master pelagic model configuration; references all other input files
`PELAGIC_MODEL_OPTIONS.txt`	Core	Model on/off switches and feature toggles
`WCONST_04.txt`	Constants	323 calibrated model constants (recommended set)
`EXTRA_WCONST.txt`	Constants	Additional constants for extended features
`INIT_CONC_1.txt`	Initial Cond.	Initial state variable concentrations (set 1)
`INIT_CONC_2.txt`	Initial Cond.	Initial state variable concentrations (set 2)
`ADVECTIVE_LINKS.txt`	Geometry	Advective transport links between boxes
`DISPERSIVE_LINKS.txt`	Geometry	Dispersive transport links between boxes

File(s)	Category	Description
BATHYMETRY_1.txt – BATHYMETRY_25.txt	Geometry	Depth profiles for each of the 25 boxes
TEMP_TS.txt	Forcing	Water temperature timeseries (all boxes)
SALT_TS.txt	Forcing	Salinity timeseries
FLOW_TS.txt	Forcing	Water flow rates
SOLAR_RAD_TS.txt	Forcing	Solar radiation timeseries
WIND_SPEED_TS.txt	Forcing	Wind speed timeseries
AIR_TEMP_TS.txt	Forcing	Air temperature timeseries
CLOUD_COVER_TS.txt	Forcing	Cloud cover timeseries
EVAPORATION_TS.txt	Forcing	Evaporation timeseries
BOUNDARY_FLOW_TS.txt	Forcing	Boundary flow timeseries
FORC_TS_1.txt – FORC_TS_9.txt	Forcing	Boundary concentration forcing timeseries
SETTLING_VELOCITY_TS_1.txt – _6.txt	Forcing	Variable settling velocity timeseries
SHEAR_STRESSES_TS.txt	Forcing	Bottom shear stress timeseries
ICE_COVER.txt	Forcing	Ice cover timeseries
ALLELOPATHIC_INFORMATION.txt	Config	Allelopathic interaction parameters
PRESCRIBED_SEDIMENT_FLUXES.txt	Sediment	Prescribed benthic sediment fluxes
PRESCRIBED_SEDIMENT_FLUXES_HYPOXIA.txt	Sediment	Hypoxic sediment flux set
FLUXES_FOR_MUDDY_SEDIMENTS*.txt	Sediment	Muddy sediment flux data (various sets)
FLUXES_FOR_SANDY_SEDIMENTS*.txt	Sediment	Sandy sediment flux data
PELAGIC_OUTPUT_INFORMATION_FILE.txt	Output	Controls which boxes produce output
RESUSPENSION_INPUTS_2.txt	Sediment	Resuspension configuration

Output Files (OUTPUTS_30day/)

After a successful 30-day run:

File	Content
PELAGIC_BOX_00005.out	Box 5 state variable timeseries
PELAGIC_BOX_00006.out	Box 6 state variable timeseries
PELAGIC_BOX_00008.out	Box 8 state variable timeseries
PELAGIC_BOX_00009.out	Box 9 state variable timeseries
PELAGIC_BOX_00014.out	Box 14 state variable timeseries
PELAGIC_BOX_00017.out	Box 17 state variable timeseries
PELAGIC_BOX_00025.out	Box 25 state variable timeseries
PELAGIC_BOX_*.mtrx	Process rate matrices (if enabled)
MASS_BALANCES.out	Element mass balance timeseries
WATER_LEVELS.out	Water level timeseries for all boxes

Observation Files (OBSERVATIONS/)

File	Description
`Water_column_chemistry_2015.xlsx`	Field water chemistry measurements
`2015ST14.xlsx`	Station 14 monitoring data
`PelagicProcesses.xlsx`	Measured pelagic process rates
`Growth_fixation_light_mixing.xlsx`	Growth and fixation measurements
`N2_fix_rates for model calibration.xlsx`	N2 fixation rate data
`sta1ND.dates`, `sta2VM.dates`	Station date index files

This tutorial was prepared for ESTAS-AQUABC v0.2. For model equations and scientific background, see the AQUABC Reference Manual *and* ESTAS Reference Manual.